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Tidal currents, sampling effort and baited remote underwater video (BRUV) surveys: Are we drawing the right conclusions?

Matthew D. Taylor[a,b,*], Jessica Baker[b,c], Iain M. Suthers[b,c]

[a] Wild Fisheries Research, Port Stephens Fisheries Institute, Locked Bag 1, Nelson Bay, New South Wales 2315, Australia
[b] Evolution and Ecology Research Centre, School of Biological, Earth and Environmental Science, University of New South Wales, New South Wales 2052, Australia
[c] Sydney Institute of Marine Science, Building 22, Chowder Bay Road, Mosman, New South Wales 2088, Australia

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ABSTRACT

Estuaries are hydrographically dynamic environments, and such variability can affect the distribution and abundance of estuarine fish. Baited remote underwater video (BRUV) can be used to quantify estuarine species, but BRUV-derived data may be confounded by variable bait plume area and the associated effect on relative sampling effort. This study investigated the potential effects of current velocity on estuarine fish abundance data, and whether associated changes in bait plume size are important for benthic BRUV surveys in estuaries. BRUV sampling was conducted across two zones in two adjacent estuaries, and current velocity measured with a drogue during each BRUV deployment. Current velocity ranged from 0.02 to 0.65 m s^{-1}, resulting in potential bait plume areas that varied by orders of magnitude. The maximum number of each species (*MaxN*) was processed to produce a standardised (by bait plume area) and unstandardised multivariate species data set. The two data sets, whilst developed from identical video footage, yielded contrasting results. Unstandardised data was more variable, but produced a stronger correlation between abiotic variables and community structure. In addition, repeated sampling at some sites revealed significant temporal variance in community structure when data was standardised by bait plume area. Variability in sampling effort resulting from variable current velocity and associated bait plume area may confound interpretation of BRUV data.

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1. Introduction

Fish community composition in estuaries can be sampled using a variety of techniques, ranging from observation to extraction. The choice of sampling methods is dependent on the focus of the study, but bias associated with each technique must be considered. A relatively new technique which is rapidly growing in application involves the use of baited remote underwater video (BRUV). BRUV surveys are non-invasive, and have the advantage of removing biases associated with divers on visual surveys (Lowry et al., 2012). The BRUV technique may also be used in habitats, or at depths which are difficult for divers to access (Willis et al., 2000), and is cost-effective for surveys which require many hours of observation (Harvey et al., 2007). In addition, the video footage obtained from BRUV surveys affords researchers the opportunity to observe fish in their natural environment (Jury et al., 2001; Watson et al., 2005), and also to re-analyse archival video footage at a later date to answer new research questions.

The application of BRUV surveys in marine ecology is increasing in frequency as technological advancements and cost reduction have made the technique both accessible and affordable (Harvey et al., 2007). BRUV surveys are often applied to examine fish populations in marine reserves (Cappo et al., 2004; McKinley et al., 2011) where extractive sampling is not ideal, and also in tropical and temperate reef (Watson et al., 2005) and pelagic environments (Heagney et al., 2007). A key element of BRUV is the use of "bait" to attract fish into the field-of-view of the video camera. The type of bait varies between studies, and can be formulated to attract a wide range of species, or can be highly specific to attract a specific species of fish. Typically, the bait disperses into the water column to form a plume of attractant down-current of the BRUV, and acts as an olfactory stimulus to attract fish. Fish follow the plume up-current to the source where their image is captured by the video camera; however, fish species that are indifferent to the plume can be opportunistically filmed as well if they wander into the field of view. Secondary attraction of larger predators outside the plume to the BRUV unit can also occur, as a result of the aggregation of smaller fishes in a small area.

There are several biases associated with traditional techniques of fish census, such as size and species selectivity when using extractive sampling using nets, and diver avoidance when using underwater visual census (Birt et al., 2012; Lowry et al., 2012).

* Corresponding author at: Wild Fisheries Research, Port Stephens Fisheries Institute, Locked Bag 1, Nelson Bay, New South Wales 2315, Australia.
Tel.: +61 249163937; fax: +61 249822265.
E-mail address: matt.taylor@dpi.nsw.gov.au (M.D. Taylor).

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Cappo et al. (2004) suggested that BRUV surveys incorporate many of the sampling advantages offered by these traditional methods, whilst avoiding some of the biases, however this and subsequent studies have illustrated several sources of potential bias in BRUV surveys. Such bias may include (1) selection for daytime BRUV deployments which can under-sample nocturnally active species (Harvey et al., 2012a); (2) selectivity by each particular bait type or formulation for only a subset of the fish community (Dorman et al., 2012); (3) the confounding effect of short-term temporal variability in fish populations (Birt et al., 2012); (4) a general negative relationship between light level (i.e. at depth or in turbid water) and fish abundance (Watson et al., 2005); (5) interspecific interactions, especially between abundant and rare species (e.g. Birt et al., 2012); and (6) the upper limit to the number of fish which can be accurately counted in the field of view (Wraith, 2007). The widespread application of BRUV surveys is now leading to an expanded literature which explicitly and quantitatively investigates the relative effect of these biases and implications for BRUV-derived data (e.g. Birt et al., 2012; Harvey et al., 2012a,b).

One particular bias involves the standardisation of sampling effort among BRUV deployments. Many organisms employ a phased strategy during olfactory-guided searches for prey or bait (e.g. Lokkeborg et al., 2000; Stiansen et al., 2010). Fish typically employ two phases of behaviour in relation to olfactory stimuli; plume-search (search for relevant stimuli), followed by the bait-search (search when an olfactory stimulus has been encountered Vabø et al., 2004). Variable bait plume size affects the probability that a fish will encounter the relevant stimuli or attractant, and thus affects the ability of any baited apparatus to sample fish or invertebrates; including fish traps (Stiansen et al., 2010), baited nets (Hill and Wassenberg, 1999), and BRUV (Heagney et al., 2007). Many BRUV studies simply hold the duration of video footage constant, or where duration differs, use time to standardise measurements taken during different deployments. Implicit in this is the assumption that each deployment samples an equivalent area of water per unit time (e.g. Hill and Wassenberg, 1999; Jones et al., 2003); however, this assumption may not be reasonable in all environments. Heagney et al. (2007) described in detail the measurement and application of bait-plume size when deploying BRUV units in often turbulent pelagic configuration in coastal areas, however few studies have yet applied this approach to either benthic or pelagic BRUV surveys. Current velocity at the bottom of the water column can vary substantially as a result of stress at the sediment–water interface (Thomas and Schallenberg, 2008). The effect of this variability on bait plume penetration and dispersal, and the importance for fish abundance needs quantitative assessment.

This study uses benthic BRUV surveys to investigate the potential effects of tidal current velocity on estuarine fish abundance data, and whether consideration of the effect of current velocity and thus associated changes in bait plume size are an important consideration for benthic BRUV surveys. Specifically, we aim to assess the effect of standardisation by current velocity and associated plume area on (1) interpretation of analyses of fish community structure using standard multivariate techniques; (2) temporal changes in fish community structure.

2. Materials and methods

2.1. Study sites and experimental design

The study was conducted in Botany Bay/Georges River estuary (34.01°S, 151.24°E) and Port Hacking (33.83°S, 151.28°E), two adjacent estuaries in the Sydney metropolitan area. Botany Bay/Georges River estuary (subsequently referred to as 'Botany Bay') is a heavily urbanised drowned river valley feeding into a coastal embayment of waterway area 92 km^2. Port Hacking is a drowned river valley of approximately 180 km^2 and a waterway area of 11 km^2. Both Port Hacking and Botany Bay are recreational fishing reserves, and contain mangrove, saltmarsh and seagrass beds.

BRUVs were deployed following a typical experimental design to examine the difference between estuarine fish assemblages within different hydrographic zones (e.g. McKinley et al., 2011). Specifically, the design included three factors: (1) zone (2-levels, Inner and Outer, fixed); (2) estuary (2-levels, Botany Bay and Port Hacking, random); and (3) site (random, nested within Zone). Estuary was included as a random factor to explore generality in any differences between Zones. Site was included as a random factor to partition variability across the spatial scale of each Zone. Zones were defined primarily on the basis of salinity. Within the 'Outer' zone, the dominant salinity influence was the tidal flows from the open ocean into the mouth of the estuary. The salinity influence of the 'Inner' zone was a combination of tides and freshwater influxes from the land including rivers and tributaries which feed into the estuary. Ten sites (~0.25 km^2) approximately 2 km apart, were selected within each zone and surveyed, with two single-camera BRUVs deployed within each site (Fig. 1).

A second, smaller experiment was also completed to assess aim 2, and evaluate the effect of standardisation on interpretation of temporal changes in fish community structure. Five sites (Site, 5-levels, fixed) within the outer zone of Port Hacking were randomly sampled at four times (Time, 4-levels, random) spread over multiple days, to capture variation in tidal and diel conditions. Two BRUV drops were simultaneously performed at random locations within each Site, at each time point.

2.2. Equipment and data collection

Each BRUV apparatus consisted of a digital video camera (Sony DCR-HC21E) mounted inside a custom-built PVC housing. The camera housing was attached to a mooring and float, and suspended ~1 m above the mooring to avoid potential interference from the epi-benthic layer of phytoplankton often present in estuaries. The bait arm extended horizontally from the camera housing, and a 13 cm × 8 cm bait housing was attached to the end of the arm exactly 1 m from the camera lens. The camera and bait arm were free to swing in the direction of the current, thus allowing fish to be observed swimming upcurrent into the bait plume. The bait housing contained a 100 g mixture of minced pilchards (80 g), falafel (10 g) and tuna oil (10 g, Folpp et al., 2011; Heagney et al., 2007; Lowry et al., 2012). Previous studies have indicated that this standardised mixture provided a constant rate of dissolution over the deployment times under a variety of conditions (Lowry et al., 2012).

Deployments across all sites were undertaken over a period of 10 deployment days during the period 22 March–4 April 2010, between 08:00 and 18:00 h. Deployments were performed in a random order, thus there was no correlation between the order in which sites were sampled, and tide or time of day. Deployments typically occurred in 5–10 m of water, and BRUVs were retrieved within 60 min of deployment. Water temperature, salinity and turbidity were recorded for each deployment using a YEO-KAL Model 611 Water Quality Analyser, with measurements taken at the same depth as the camera housing.

Water current data was collected using a drifting drogue (as described in Heagney et al., 2007). We used the drogue drift rates to determine current velocity and estimate plume area. A drifting drogue only represents two dimensions of the three dimensional advection of the plume, and does not account for turbulence and dispersion, but despite these potential limitations we assume that this relatively easy approach to collecting current velocity data to

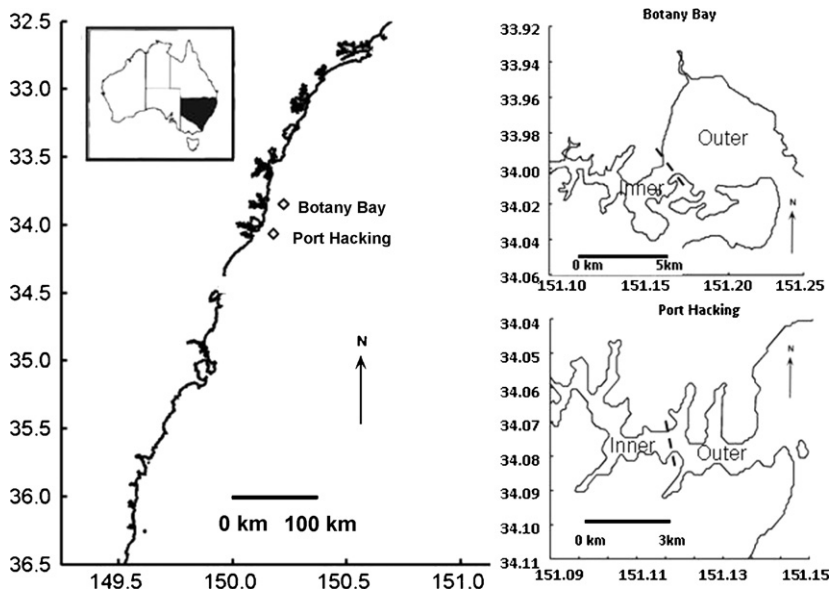

Fig. 1. Chart of Botany Bay and Port Hacking, two estuaries located on the mid-NSW coast. Zones, as described in Section 2.1, are indicated for each estuary.

measure BRUV effort in a tidal estuary and is better than the usual assumption of no variation in sampling effort. The drogue was constructed from interconnecting wooden panels that were weighted and suspended at approximately the same depth as the BRUV unit, and attached to 10 m of graduated rope. The drogue was dropped from an anchored vessel and the time taken for the drogue to drift for the length of the 10 m rope recorded. In areas of low current velocity, the drogue was allowed to drift for 1 min, and the distance recorded. The distance and time measurements were used to calculate the sub-surface current velocity, which was used to calculate the sampling "effort" associated with each BRUV deployment (described below).

2.3. Image and data analysis

Digital imagery recorded by the camera was downloaded directly to a PC using the video capturing software supplied with the camera and stored in its original resolution as a .mov file on a 1.5TB external hard drive. The PC was connected to a 42 in. Sony Bravia LCD television via a VGA cable and videos were then reviewed using VLC Media Player v. 1.1.11. Videos were selected from the external hard drive for analysis in a random order, and the first 45 min of footage analysed by the same reader. The length of the bait arm (1 m) was used as a guide to estimate capture-range directly from the footage, and fish observed outside this range were excluded from the data count to allow accurate species identification and control for the effect of visibility. Each taxa observed in the video was identified to species, and a specialist was consulted to confirm the correct identification of some species where required. Species data was expressed as *MaxN* (Ellis and Demartini, 1995), the maximum number of fish of a particular species present in a single video frame during the first 45 min of video footage (thus following the approach outlined in Heagney et al., 2007). The species data derived from each drop was also converted into an estimate of abundance standardised by a measure of effort using a simple triangular bait plume model. We acknowledge this is an overly simple approach, however in the absence of any other information of bait plume dynamics in a tidal estuary it is a suitable estimate of effort in this study (but deserves further research). Standardisation was achieved by using current velocity to calculate the area covered by the bait plume, as observed in the denominator of the following equation (Heagney et al., 2007; Hill and Wassenberg, 1999; Sainte-Marie and Hargrave, 1987):

$$MaxN_{Area} = \frac{MaxN}{0.165 \cdot (t_d \cdot v_w)^2}$$

where $MaxN_{Area}$ is *MaxN* standardised by an approximation of a triangular area bait plume (fish m^{-2}), t_d is the time of the deployment (in s; i.e. 2700 s in each 45 min deployment), and v_w is the current velocity measured during the deployment ($m\,s^{-1}$). This yielded two data sets reflecting non-standardised (*MaxN*) and standardised data ($MaxN_{Area}$; note that throughout this manuscript "standardised" refers only to standardisation by the bait plume, not standardisation as outlined in Anderson et al., 2008).

Correlations between abiotic (depth, temperature, salinity, turbidity, time-of-day, and also current velocity) were evaluated using the *Draftsman's Plot* function in PRIMER 6 (PRIMER-E, Plymouth). This operation plots all pairs of variables against each other, and also computes the Spearman correlation between each pair of variables to provide an estimate of any interrelationships (Clarke and Ainsworth, 1993). Estimated plume area was compared among tidal period (4-levels, Incoming, Outgoing, Slack high, Slack low, fixed) using PERMANOVA (Anderson, 2001). Depth, temperature and turbidity were normalised and used to calculate a Euclidian distance matrix among samples, and differences among Site, Estuary, and Zone evaluated using PERMANOVA. Species assemblage data were $\log_{10}(x+1)$ transformed to emphasise rare species which are more likely to define individual assemblages, and sample resemblance expressed using a Bray–Curtis similarity matrix. A non-metric multidimensional scaling (nMDS) ordination was produced from the sample resemblance matrix for both *MaxN* and $MaxN_{Area}$, to visually compare the effects of standardisation on the centroid and spread of multivariate data in 2-dimensional space. Stress values were compared to evaluate how well the 2-dimensional ordination represented the multivariate data. Species data for both *MaxN* and $MaxN_{Area}$ data sets was also analysed using PERMANOVA and the design described in Section 2.1. A further analysis was also undertaken, comparing *MaxN* using the design described in Section 2.1, but including plume area as a covariate. The contribution of individual species to patterns revealed using PERMANOVA was examined using a SIMPER analysis.

Table 1
Taxonomic classification of species observed during BRUV surveys in Port Hacking and Botany Bay. Estuary and Zone indicate the estuaries (PH: Port Hacking; BB: Botany Bay) and zones each species was observed in.

Family	Species	Common name	Estuary	Zone
Order: Myliobatiformes				
Urolophidae	*Trygonoptera testacea*	Common stingaree	PH	Inner
Order: Perciformes				
Carangidae	*Trachurus novaezelandiae*	Yellowtail scad	PH, BB	Outer
	Pseudocaranx dentex	Silver trevally	PH, BB	Outer
Cheilodactylidae	*Nemadactylus macropterus*	Jackass morwong	BB	Outer
	Cheilodactylus fuscus	Red morwong	PH	Outer
Dinolestidae	*Dinolestes lewini*	Long-fin pike	PH	Outer
Kyphosidae	*Girella tricuspidata*	Luderick	PH	Inner
	Girella elevata	Rock blackfish	PH	Inner
	Atypichthys strigatus	Mado	PH	Outer, Inner
Pomatomidae	*Pomatomus saltatrix*	Tailor	PH	Inner
Sillaginidae	*Sillago cilliata*	Sand whiting	PH	Outer, Inner
Sparidae	*Acanthopagrus australis*	Yellowfin bream	PH, BB	Outer, Inner
	Rhabdosargus sarba	Tarwhine	BB	Outer, Inner
	Pagrus auratus	Snapper	PH	Outer, Inner
Order: Tetraodontiformes				
Monacanthidae	*Nelusetta ayraud*	Ocean jacket	PH, BB	Outer, Inner
	Meuschenia trachylepis	Yellow-finned leatherjacket	PH, BB	Outer, Inner
	Meuschenia freycineti	Sixspine leatherjacket	PH	Outer, Inner
	Scobinichthys granulatus	Rough leatherjacket	PH	Outer, Inner
	Monacanthus chinensis	Fanbelly leatherjacket	PH	Outer, Inner
Ostraciidae	*Anoplocapros inermis*	Eastern smooth boxfish	PH	Outer, Inner
Tetraodontidae	*Torquigener pleurogramma*	Weeping toado	PH, BB	Outer

A distance-based linear model (DistLM, Anderson et al., 2006) was used to compare correlations between water temperature, turbidity and depth, and the species assemblage for both data *MaxN* and $MaxN_{Area}$ species assemblage data. This routine calculates a distance-based multivariate multiple regression for any linear model on the basis of any distance measure, using permutation procedures. The 'Best' selection procedure was employed using Akaike Information Criteria (AIC) model selection criteria. Correlations between abiotic factors and community and species data were examined using principal coordinated ordination (PCO). PCO is a PERMANOVA function that performs a principal coordinate analysis of any symmetric distance matrix. This analysis is also called metric multi-dimensional scaling (Anderson et al., 2008). All univariate and multivariate comparisons were made using PRIMER 6 with PERMANOVA+ (PRIMER-E, Plymouth).

3. Results

A total of 21 different species from three orders were observed in 110 h of BRUV footage recorded during the study (Table 1). The most abundant Order was Perciformes, for which 13 species were observed, followed by Tetraodontiformes. In Port Hacking, 19 of the 21 species were observed during the study. Conversely, only 7 out of 21 species were observed in Botany Bay. Most BRUV drops occurred over sandy substrates, with minimal aquatic vegetation.

3.1. Hydrographic variability

There was a negative correlation between turbidity and salinity (−0.69, Table 2), with only weak correlations between any of the other variable pairs (Table 2). As salinity showed greater skewness, turbidity was retained in further analyses and salinity excluded. There was no obvious correlation between current velocity and any of the other abiotic factors (−0.18 to 0.07, Table 2). Benthic current velocity ranged between 0.02 and 0.65 $m\,s^{-1}$, and estimated plume area ranged between 50 m^2 and 2280 m^2. There was no significant difference in plume area among tidal periods ($F = 1.25$, $P = 0.27$, Fig. 2) indicating that whilst current velocity approximated tidal current movement, there were extrinsic factors such as wind movement which contributed to measured current velocity and associated plume area. Average current measurements taken using the drogue in this study were similar to those observed for other estuaries with similar tidal prisms (e.g. Riedel and Gourlay, 1980).

The differences in the hydrographic properties of different Zone's and Estuary's was confirmed by PERMANOVA, with significant Zone*Estuary ($F = 7.63$, $P < 0.01$) and Estuary*Site(Zone) ($F = 8.58$, $P \ll 0.01$) interactions. These results indicate that outer zones were fairly similar for both estuaries, but inner zones differed; and also that there was significant variability in water quality

Table 2
Spearman correlations between continuous environmental variables measured alongside BRUV deployments.

	Time of deployment	Temperature	Salinity	Turbidity	Current velocity
Temperature	0.27				
Salinity	0.09	0.03			
Turbidity	−0.20	−0.23	−0.69		
Current velocity	−0.18	0.07	−0.05	0.02	
Depth	0.03	0.10	0.25	−0.14	−0.12

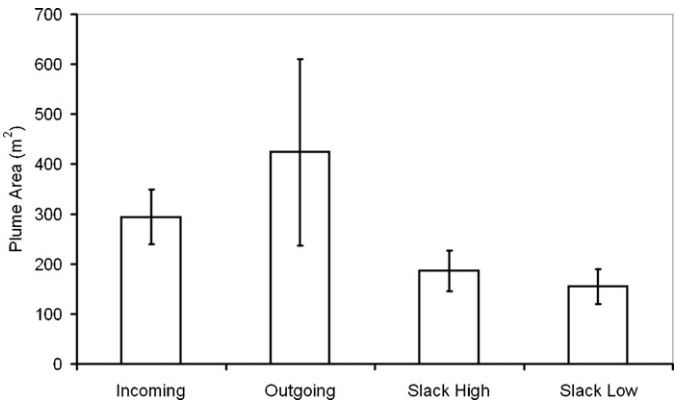

Fig. 2. Mean ± SE bait plume area estimated from current velocity, and grouped by tidal period. Plume area ranged from 50 m^2 to 2280 m^2.

among sites. As expected, benthic current velocity was significantly different among Site(Zone) ($F = 30.39$, $P \ll 0.01$), however there were no consistent differences between Zone's or Estuary's.

3.2. Multivariate analysis of MaxN and $MaxN_{Area}$

Non-metric multidimensional scaling indicated distinct ordinations for *MaxN* and $MaxN_{Area}$ (Fig. 3a and b), and this was statistically confirmed by PERMANOVA. The 2-dimensional stress values (Fig. 3a and b) indicated that the $MaxN_{Area}$ could be more easily represented in a 2-dimensional ordination than *MaxN*. PERMANOVA revealed significant interaction terms, with

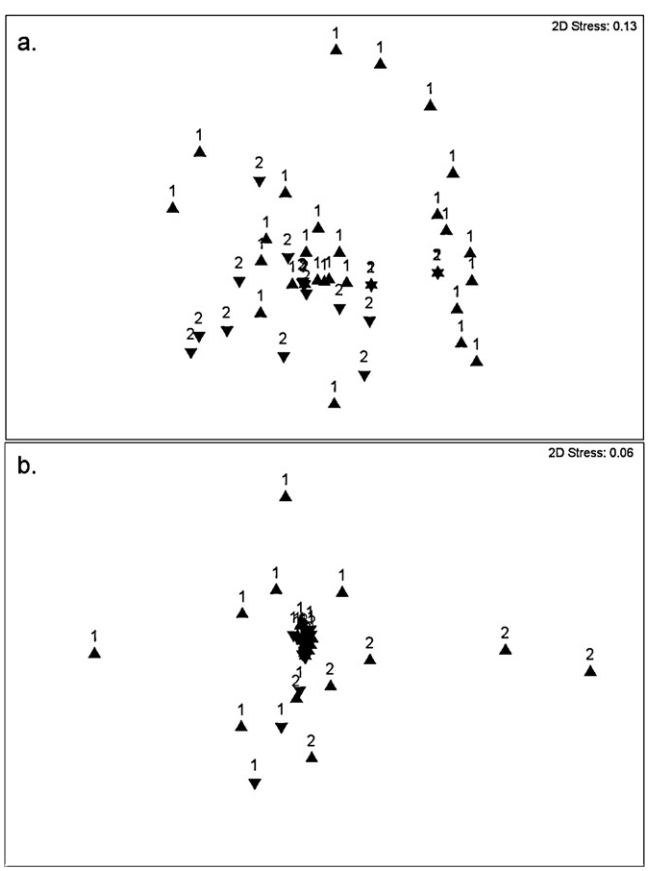

Fig. 3. Non-metric multidimensional scaling ordination of community *MaxN* (a) and $MaxN_{Area}$ (b) data, showing distinct ordinations for each data set. Data series indicate Port Hacking (▲) and Botany Bay (▼), and Outer (1) or Inner (2) zones in both figure panels.

Table 3
Species contributions (SIMPER) to differences observed between zones, for both *MaxN* and $MaxN_{Area}$. Numbers reflect percentages, and species for which the order of contribution differs between *MaxN* and $MaxN_{Area}$ are indicated with an asterisk (*). The total contribution to dissimilarity between zones for the species listed was 91% for *MaxN* and 90% for $MaxN_{Area}$.

Species	*MaxN*	$MaxN_{Area}$
Yellowfin bream	45.0	46.7
Ocean jacket	10.7	10.4
Eastern smooth boxfish	8.3	7.9
Silver trevally	6.8	6.8
Tarwhine	5.3	4.0*
Mado	4.6	6.6*
Rough leatherjacket	2.6	2.3*
Yellowtail scad	2.5	3.3*
Highfin amberjack	2.1	2.1*
Weeping toado	1.8	
Snapper	1.8	

Estuary*Site(Zone) interaction for both *MaxN* ($F_{18,40} = 1.72$, $P < 0.01$) and $MaxN_{Area}$ ($F_{18,40} = 1.54$, $P = 0.01$); and a significant Zone*Estuary interaction for both *MaxN* ($F_{1,40} = 3.95$, $P = 0.03$) and $MaxN_{Area}$ ($F_{1,40} = 2.91$, $P = 0.04$). Pairwise comparisons (PERMANOVA) were used to further explore the Zone*Estuary interaction term for each data set, by comparing levels of Zone separately for each level of Estuary. Pairwise comparisons revealed differences in community structure calculated using *MaxN* between zones for Botany Bay ($t = 2.39$, $P = 0.01$) but not Port Hacking. Conversely, the community structure calculated using $MaxN_{Area}$ was different between zones for Port Hacking only, and not Botany Bay ($t = 1.63$, $P < 0.01$). When Plume area was included as a covariate in the analysis of *MaxN*, the significant Zone*Estuary interaction remained ($F_{1,39} = 5.00$, $P < 0.01$), however the there was also a significant Plume area*Site(Zone) interaction ($F_{1,39} = 1.73$, $P < 0.01$). Pairwise comparisons of the Zone*Estuary interaction term showed the same relationships as for *MaxN* without the covariate.

The differences between zones were primarily driven by yellowfin bream, followed by ocean jacket, eastern smooth boxfish, and silver trevally (SIMPER analysis, Table 3). Other species that contributed to the observed trends were similar between data sets, however the relative percent contribution of these species differed between *MaxN* and $MaxN_{Area}$ (Table 3). The potential contribution of different fish behaviour across different tides to our results was examined by graphing relative *MaxN* over different tidal periods for yellowfin bream, ocean jacket and silver trevally (Fig. 4). These results showed minor differences across tides, although the *MaxN* of ocean jackets was noticeably higher during the slack-high tide. Overall, sample variation was much lower for the $MaxN_{Area}$ data set, as observed in the coefficient-of-variation values given in Fig. 4.

DistLM analysis revealed correlations between several variables and both *MaxN* and $MaxN_{Area}$ multivariate community structure. For *MaxN*, the best model (AIC = 561.4, $R^2 = 0.17$) reflected a significant correlation between community structure and temperature ($F = 4.91$, $P < 0.01$), turbidity ($F = 5.25$, $P < 0.01$), and water depth ($F = 3.00$, $P = 0.03$; Fig. 5a). PCO plots did not indicate clear clustering along the PCO axes for *MaxN* (Fig. 5a). Fig. 5a also presents species which were correlated (Pearson correlation factor >0.65) with first two PCO axes, on the basis of *MaxN*. Common species yellowfin bream, silver trevally and ocean jacket all contributed to the observed differences, with the latter two species being more abundant in Outer zone sites.

For $MaxN_{Area}$, the best model represented a poor model fit (AIC = 230.2, $R^2 = 0.05$), but did reflect a significant correlation between the observed community structure and temperature ($F = 3.60$, $P < 0.01$, Fig. 5b). Again, there was no clear clustering along the PCO axes for $MaxN_{Area}$ (Fig. 5b); although PCO1 appeared to describe some separation between zones, and some species were correlated (Pearson correlation factor >0.2) with the calculated PCO

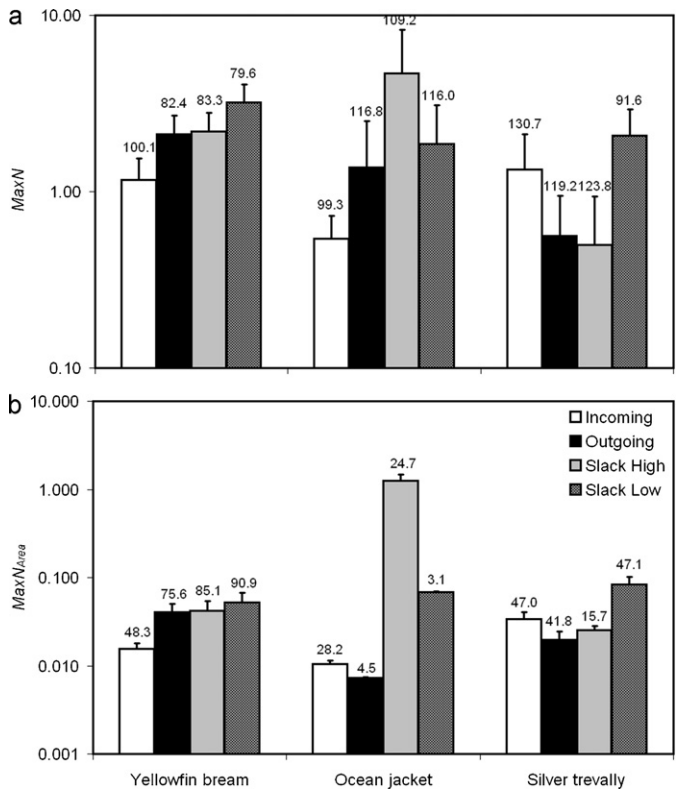

Fig. 4. Mean ± SE *MaxN* (a) and $MaxN_{Area}$ (b) of the three most important species observed in the current study, grouped by tidal period. Coefficients-of-variation are included above each error bar to show the relative variability of each measurement in the *MaxN* and $MaxN_{Area}$ data sets.

axes (including common species yellowfin bream and mado, and rare species including sand whiting and common stingaree). The distributions of these species did not approximate any vector lines in Fig. 5b, meaning there were no obvious over-arching correlations between abiotic variables and particular species important in driving trends in the multivariate assemblage when data were adjusted for bait plume area.

3.3. Temporal changes in fish community structure determined using MaxN and $MaxN_{Area}$

When compared using PERMANOVA, community data calculated using *MaxN* indicated there were no significant differences among levels within factor Time ($F_{3,20} = 1.34$, $P = 0.09$). This is in contrast to analysis of community data calculated using $MaxN_{Area}$, where a significant difference was detected within Time ($F_{3,20} = 2.97$, $P \ll 0.01$). There were no significant differences between Site for either *MaxN* ($F_{3,20} = 1.08$, $P = 0.38$) or $MaxN_{Area}$ ($F_{3,20} = 1.09$, $P = 0.18$).

4. Discussion

4.1. Analytical considerations and inferences

The application of the BRUV technique in estuarine and coastal waters, with tidal and site specific currents, can potentially result in different conclusions if the effect of variable sampling effort is considered. This finding is important, particularly where data derived from the survey method may directly contribute to major management decisions in coastal, tidal waters, such as within marine protected areas (e.g. Cappo et al., 2004). Whilst many sources of potential bias are situation specific (e.g. Broad et al., 2010), the potential effect of current velocity on relative effort is relevant across a wide spectrum of environments. For example, current velocity and effective swimming speed are essential to consider in deep-sea BRUV (Priede and Merrett, 1996), but not for pelagic BRUV (Heagney et al., 2007); and the current study indicates it is important to at least consider these factors when conducting BRUV surveys in estuaries.

The drogue and triangular bait plume approach adopted in this study represents a simple, but effective way of assessing relative current velocity. An alternative approach would be to attach individual current metres to each BRUV unit, but this would come at added expense to the construction of each unit. We assumed a measure of effort as a triangular bait plume size based on the drogue's drift, as our BRUV was similar to that deployed by Heagney et al. (2007); it was 1 m off the flat estuary floor, and was free to rotate and align with the tidal current, pointing down-current. Theoretically, the plume has a 3-dimensional volume which would be site and tide-specific, and would involve relatively complex hydrodynamic modelling to characterise. Further, in

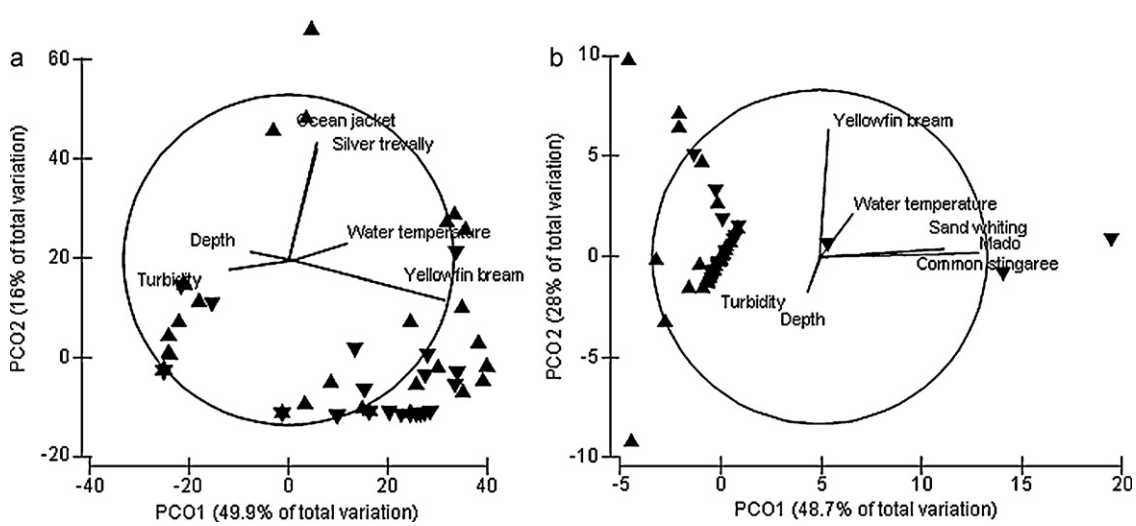

Fig. 5. Principle coordinated (metric multidimensional scaling) ordination of community *MaxN* (a) and $MaxN_{Area}$ (b) data, showing correlations between Outer (▲) and Inner (▼) zones and PCO axes 1 and 2. Loadings of abiotic and species data are shown for Pearson correlations >0.65.

some situations the effects of turbulence from benthic features could greatly increase the volume of the bait plume, although our deployments were not near rocky reefs. There is clearly a need for further research in this area, particular in dynamic hydrographic environments like estuaries. A greater understanding of bait plume dynamics in estuaries will improve the way such information can be incorporated into future analyses.

Our data indicate that within estuarine environments, variable epibenthic current velocity has the potential to alter the conclusions drawn from studies of fish communities. To illustrate this, consider in simple terms the conclusions that may be drawn from results of the analysis of *MaxN* and $MaxN_{Area}$. The results of both PERMANOVA support the conclusion that fish community structure varies among different hydrographic zones, but the effect is estuary specific. Potential bias associated with current velocity and associated changes to bait plume become evident when the effect is considered through pairwise comparisons. On the basis of *MaxN* data, the conclusion would be that there was a difference between zones in Botany Bay and not Port Hacking. Conversely, on the basis of $MaxN_{Area}$ data the conclusion would be that there was a difference between zones in Port Hacking and not Botany Bay. The complex effects of bait plume dynamics became more evident when Plume area was included as a covariate in the analysis of *MaxN*, where a significant Plume Area*Site(Zone) interaction implied that the effects of Plume area were site-specific. Ultimately, the interpretation of results was altered when data were standardised by plume area, and potential reasons for this are discussed later (such as hydrographic variability and fish behaviour). The overall variability in the community data was reduced with standardisation, and much of this variation was explained by Plume area (Fig. 4). Consequently, standardising by plume area may reduce the overall variability in BRUV samples, and may ultimately reduce the sample size required for adequate statistical power of multivariate comparisons (Irvine et al., 2011 present a power analysis protocol for multivariate tests).

The calculation of various indices for comparisons of community diversity among different areas is frequently performed using BRUV data (e.g. Colton and Swearer, 2010; Lowry et al., 2011; Wraith, 2007). In particular, these studies mostly calculate relative measures of diversity, such as the Shannon Index (Shannon and Weaver, 1963) and evenness (Pielou, 1966). Despite the effect of standardisation by bait plume area on relative abundance estimates discussed above, calculation of relative measures of diversity remain unaffected by sampling effort. This is due to the fact that the proportional contribution of species within a sample (i.e. a particular BRUV deployment) remains the same, regardless of any standardisation, since all species within a sample are standardised by the same estimate of bait plume area.

Logistical constraints experienced in BRUV studies (such as time, daylight, and the number of BRUV units available) mean spatial comparisons of fish abundances may be confounded by short-term temporal variations in weather, time of day, tidal, or lunar effects (Westera et al., 2003). A common solution to this problem is to temporally randomise BRUV deployments at different sites, such that there is no correlation between time of deployment and site or zone. Westera et al. (2003) tested short-term variability by repeatedly sampling four sites with BRUV surveys, 3 days apart, and did not find any significant difference in the fish assemblage among days. Our study showed that whilst there was no difference among random time points within sites for *MaxN*, there were significant differences for $MaxN_{Area}$. This supports the idea that variable bait plume area may indeed result in a large amount of variation in multivariate BRUV data sets, and this could contribute to a lack of significant differences among samples or days. The temporal comparison illustrated here indicates that this variation is sufficient to produce a Type II error (which in this case is incorrectly retaining the null hypothesis of *no difference in community structure among time points*). This has implications for BRUV studies which either use temporally randomised sampling to remove the effect of abiotic variability, or include a random temporal factor in the overall experimental design to partition this variation in the analysis. To improve confidence in inferences made from statistical analyses of BRUV data, studies should at least consider standardisation of data using bait plume area or some other measure of effort associated with bait dispersal.

4.2. Hydrographic and behavioural considerations

Distance-based linear modelling (DistLM) of *MaxN* and $MaxN_{Area}$ data sets produced inconsistent relationships. The measured abiotic variables did not explain a great deal of the variation in observed community structure, however in a relative sense the amount of variation explained by abiotic variables was much greater for *MaxN* than $MaxN_{Area}$. In addition, different species produced strong correlations with the PCO axes between the data sets. Relatively rarer species correlated with the PCO axes for $MaxN_{Area}$ data sets, but there were no over-arching correlations between particular species and abiotic variables.

Although the results of the DistLM analyses are largely inconclusive, the nature of correlations between community structure and abiotic variables is obviously affected by standardisation.

In estuaries, variation in water quality variables can be correlated with the tide (e.g. DiLorenzo et al., 2004; Nezlin et al., 2009), as can current velocity and estimates of bait plume area. Consequently, bait plume area could be expected to explain much of the variation in community structure. This creates an interesting paradox when deciding whether to standardise by bait plume area when analysing data. Because current velocity can co-vary with many other abiotic variables which change alongside tide, standardising could adversely affect the ability of analyses such as DistLM to resolve relationships between abiotic variables and community structure. If data are not standardised, however, researchers run the risk of incorrectly concluding that correlations exist between community structure and particular abiotic variables, when in reality the correlation might simply be due to differences in sampling effort associated with variable water movement. A potential solution to this when dealing with continuous variables is to use unstandardised data, but including plume area as an additional covariate in the analyses to partition some of variation associated with sampling effort alongside the other covariates.

In our analyses, however, temperature did form a significant correlation with community structure, which is to be expected based on existing knowledge of the species driving the observed correlations. Many past studies have observed a significant correlation between community composition and water temperature (Loneragan et al., 1986; Whitfield, 1994), and in our study the significance of water temperature is likely linked to the biology of yellowfin bream. Yellowfin bream are catadromous and tolerant to environmental fluctuations (Kroon, 2005; Thomas and Connolly, 2001). This species is usually prevalent in warmer water temperatures, which has associated benefits for growth (Griffiths, 2001) and yet are relatively indifferent to other abiotic variables, such as turbidity and salinity (Cyrus and Blaber, 1987).

There are several additional factors which may co-vary with water currents, and could also contribute to observed variation in fish community structure. For example, differences in plume direction may sample a different set of habitats during different tides, or fish may behave differently and be more susceptible to BRUV sampling during different tides (e.g. Birt et al., 2012). For example, Fig. 4 indicates that there is some variation in how fishes were sampled during different tides in our study, particularly for ocean jackets. Whilst there is no specific information regarding the swimming

ability of *Meuschenia trachylepis*, evidence suggests that Monacanthidae are relatively poor swimmers (Fisher et al., 2005), and this may reduce their ability to travel up the bait plume during different tidal or current conditions. Effective swimming speed of the fish and home range size of the individual species may have direct effects on the observed fish community, and it is important to consider this as some species may be under- or over-sampled during different tidal or current conditions. Whilst there is no information on home range or swimming speed for important species in this study, the increasing acoustic telemetry literature may provide supporting information to assist in the interpretation of BRUV data in the future.

5. Conclusions

The results presented here indicate that standardisation of BRUV data by sampling effort can affect the conclusions drawn from BRUV studies. We recommend that researchers consider some measure of spatial sampling effort in BRUV studies, particularly in highly dynamic environments such as estuaries and coastal waters. Improved understanding of bait plume dynamics in estuaries will improve the ability of researchers to account for spatial variation in sampling effort. There are several other solutions which may also address the potential problems identified with variable sampling effort, such as reducing the sampling time (and thus the penetration of the bait plume), or employing underwater water video without an any bait. This will likely reduce statistical power, so there are some advantages of estimating and standardising by bait plume size over these approaches. Researchers should consider the potential consequences of both standardising data and failing to standardise by some measure of effort, and place suitable caveats on results and conclusions as appropriate for their data set and analyses.

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References

Anderson, M.J., 2001. A new method for non-parametric multivariate analysis of variance. Austral. Ecol. 26, 32–46.
Anderson, M.J., Ellingsen, K.E., McArdle, B.H., 2006. Multivariate dispersion as a measure of beta diversity. Ecol. Lett. 9, 683–693.
Anderson, M.J., Gorley, R.N., Clarke, K.R., 2008. PERMANOVA+ for PRIMER: Guide to Software and Statistical Methods. PRIMER-E, Plymouth, UK, 24 pp.
Birt, M.J., Harvey, E.S., Langlois, T.J., 2012. Within and between day variability in temperate reef fish assemblages: learned response to baited video. J. Exp. Mar. Biol. Ecol. 416–417, 92–100.
Broad, A., Knott, N., Turon, X., Davis, A.R., 2010. Effects of a shark repulsion device on rocky reef fishes: no shocking outcomes. Mar. Ecol. Prog. Ser. 408, 295–298.
Cappo, M., Speare, P., De'ath, G., 2004. Comparison of baited remote underwater video stations (BRUVS) and prawn (shrimp) trawls for assessments of fish biodiversity in inter-reefal areas of the Great Barrier Reef Marine Park. J. Exp. Mar. Biol. Ecol. 302, 123–152.
Clarke, K.R., Ainsworth, M., 1993. A method of linking multivariate community structure to environmental variables. Mar. Ecol. Prog. Ser. 92, 205–219.
Colton, M.A., Swearer, S.E., 2010. A comparison of two survey methods: differences between underwater visual census and baited remote underwater video. Mar. Ecol. Prog. Ser. 400, 19–36.
Cyrus, D.P., Blaber, S.J.M., 1987. The influence of turbidity on juvenile marine fishes in estuaries. Part 1. Field studies at Lake St. Lucia on the southeastern coast of Africa. J. Exp. Mar. Biol. Ecol. 109, 53–70.
DiLorenzo, J.L., Filadelfo, R.J., Surak, C.R., Litwack, H.S., Gunawardana, V.K., Najarian, T.O., 2004. Tidal variability in the water quality of an urbanized estuary. Estuaries 27, 851–860.
Dorman, S.R., Harvey, E.S., Newman, S.J., 2012. Bait effects in sampling coral reef fish assemblages with stereo-BRUVs. PLoS ONE 7, e41538.
Ellis, D.M., Demartini, E.E., 1995. Evaluation of a video camera technique for indexing abundances of juvenile pink snapper, *Pristipomoides filamentosus*, and other Hawaiian insular shelf fishes. Fish. Bull. 93, 67–77.
Fisher, R., Leis, J., Clark, D., Wilson, S., 2005. Critical swimming speeds of late-stage coral reef fish larvae: variation within species, among species and between locations. Mar. Biol. 147, 1201–1212.
Folpp, H., Lowry, M., Gregson, M.A., Suthers, I.M., 2011. Colonisation and community development of fish assemblages assciated with estuarine artificial reefs. Braz. J. Oceanogr. 59, 55–67.
Griffiths, S., 2001. Factors influencing fish composition in an Australian intermittently open estuary. Is stability salinity dependent? Estuarine Coastal and Shelf Science 52, 739–751.
Harvey, E.S., Cappo, M., Butler, J.J., Hall, N., Kendrick, G.A., 2007. Bait attraction affects the performance of remote underwater video stations in assessment of demersal fish community structure. Mar. Ecol. Prog. Ser. 350, 245–254.
Harvey, E.S., Butler, J.J., McLean, D.L., Shand, J., 2012a. Contrasting habitat use of diurnal and nocturnal fish assemblages in temperate Western Australia. J. Exp. Mar. Biol. Ecol. 426–427, 78–86.
Harvey, E.S., Newman, S.J., McLean, D.L., Cappo, M., Meeuwig, J.J., Skepper, C.L., 2012b. Comparison of the relative efficiencies of stereo-BRUVs and traps for sampling tropical continental shelf demersal fishes. Fish. Res. 125–126, 108–120.
Heagney, E.C., Lynch, T.P., Babcock, R.C., Suthers, I.M., 2007. Pelagic fish assemblages assessed using mid-water baited video: standardising fish counts using bait plume size. Mar. Ecol. Prog. Ser. 350, 255–266.
Hill, B.J., Wassenberg, T.J., 1999. The response of spanner crabs (*Ranina ranina*) to tangle nets – behaviour of the crabs on the nets, probability of capture and estimated distance of attraction to bait. Fish. Res. 41, 37–46.
Irvine, K.M., Dinger, E.C., Sarr, D., 2011. A power analysis for multivariate tests of temporal trend in species composition. Ecology 92, 1879–1886.
Jones, E.G., Tselepides, A., Bagley, P.M., Collins, M.A., Priede, I.G., 2003. Bathymetric distribution of some benthic and benthopelagic species attracted to baited cameras and traps in the deep eastern Mediterranean. Mar. Ecol. Prog. Ser. 251, 75–86.
Jury, S.H., Howell, H., O'Grady, D.F., Watson, W.H., 2001. Lobster trap video: in situ video surveillance of the behaviour of *Homarus americanus* in and around traps. Mar. Freshw. Res. 52, 1125–1132.
Kroon, F.J., 2005. Behavioural avoidance of acidified water by juveniles of four commercial fish and prawn species with migratory life stages. Mar. Ecol. Prog. Ser. 285, 193–204.
Lokkeborg, S., Skajaa, K., Ferno, A., 2000. Food-search strategy in ling (*Molva molva* L.): crepuscular activity and use of space. J. Exp. Mar. Biol. Ecol. 247, 195–208.
Loneragan, N.R., Potter, I.C., Lenanton, R.C.J., Caputi, N., 1986. Spatial and seasonal differences in the fish fauna in the shallows of a large Australian estuary. Mar. Biol. 92, 575–586.
Lowry, M., Folpp, H., Gregson, M., 2011. Evaluation of an underwater solid state memory video system with application to fish abundance and diversity studies in southeast Australia. Fish. Res. 110, 10–17.
Lowry, M., Folpp, H., Gregson, M., Suthers, I., 2012. Comparison of baited remote underwater video (BRUV) and underwater visual census (UVC) for assessment of artificial reefs in estuaries. J. Exp. Mar. Biol. Ecol. 416, 243–253.
McKinley, A.C., Ryan, L., Coleman, M.A., Knott, N.A., Clark, G., Taylor, M.D., Johnston, E.L., 2011. Putting marine sanctuaries into context: a comparison of estuary fish assemblages over multiple levels of protection and modification. Aquat. Conserv.: Mar. Freshw. Ecosys. 21, 636–648.
Nezlin, N.P., Kamer, K., Hyde, J., Stein, E.D., 2009. Dissolved oxygen dynamics in a eutrophic estuary, Upper Newport Bay, California. Estuar. Coast. Shelf Sci. 82, 139–151.
Pielou, E.C., 1966. Measurement of diversity in different types of biological collections. J. Theor. Biol. 13, 131.
Priede, I.G., Merrett, N.R., 1996. Estimation of abundance of abyssal demersal fishes; a comparison of data from trawls and baited cameras. J. Fish Biol. 49, 207–216.
Riedel, H.P., Gourlay, M.R., 1980. Inlets/estuaries discharging into sheltered waters. In: Proceedings of the 17th Coastal Engineering Conference, vol. 153, pp. 2550–2565.
Sainte-Marie, B., Hargrave, B.T., 1987. Estimation of scavenger abundance and distance of attraction to bait. Mar. Biol. 94, 431–443.
Shannon, C.E., Weaver, W., 1963. The Mathematical Theory of Communication. University of Illinois Press, Urbana, 117 pp.
Stiansen, S., Ferno, A., Furevik, D., Jorgensen, T., Lokkeborg, S., 2010. Horizontal and vertical odor plume trapping of red king crabs explains the different efficiency of top- and side-entrance pot designs. Trans. Am. Fish. Soc. 139, 483–490.
Thomas, B.E., Connolly, R.M., 2001. Fish use of subtropical saltmarshes in Queensland, Australia: relationships with vegetation, water depth and distance onto the marsh. Mar. Ecol. Prog. Ser. 209, 275–288.
Thomas, D., Schallenberg, M., 2008. Benthic shear stress gradient defines three mutually exclusive modes of non-biological internal nutrient loading in shallow lakes. Hydrobiologia 610, 1–11.
Vabø, R., Huse, G., Fernö, A., Jørgensen, T., Løkkeborg, S., Skaret, G., 2004. Simulating search behaviour of fish towards bait. ICES J. Mar. Sci. 61, 1224–1232.

Watson, D., Harvey, E., Anderson, M., Kendrick, G., 2005. A comparison of temperate reef fish assemblages recorded by three underwater stereo-video techniques. Mar. Biol. 148, 415–425.
Westera, M., Lavery, P., Hyndes, G., 2003. Differences in recreationally targeted fishes between protected and fished areas of a coral reef marine park. J. Exp. Mar. Biol. Ecol. 294, 145–168.
Whitfield, A.K., 1994. Abundance of larval and 0+ juvenile marine fishes in the lower reaches of three southern African estuaries with differing freshwater inputs. Mar. Ecol. Prog. Ser. 105, 257–267.
Willis, T., Millar, R., Babcock, R., 2000. Detection of spatial variability in relative density of fishes: comparison of visual census, angling, and baited underwater video. Mar. Ecol. Prog. Ser. 198, 249–260.
Wraith, J.A., 2007. Assessing reef fish assemblages in a temperate marine park using baited remote underwater video. School of Biological Sciences, University of Wollongong, 88 pp.